

Literature Review: Knowledge Graph-Based and Attention-Enhanced Approaches for Fake News Detection

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November 30, 2025

1 Literature Review

1.1 Knowledge Graph-Based Approaches for Fake News Detection

The integration of knowledge graphs (KGs) has emerged as a transformative approach in fake news detection, addressing the limitations of traditional text-based methods that lack factual verification capabilities. Early foundational work by [8] introduced DEAP-FAKED, which combined natural language processing with tensor decomposition to encode both news content and KG entities. Their framework achieved F1-scores of 88% and 78% on

two benchmark datasets, demonstrating the potential of KG integration for improving detection accuracy.

Building upon this foundation, recent research has explored more sophisticated KG architectures. [10] proposed KEHGNN-FD, a knowledge graph-enhanced heterogeneous graph neural network that models news, topics, and entities within a unified framework. Their approach outperformed seven baseline methods by effectively aggregating information through attention mechanisms. Similarly, [3] developed KG-MFEND, an efficient knowledge graph-based model for multi-domain fake news detection that constructs comprehensive multi-domain KGs to enhance generalization capabilities across different contexts.

1.2 Attention Mechanisms and Graph Neural Networks

The incorporation of attention mechanisms has significantly advanced the field by enabling models to focus on the most relevant information for fake news detection. [5] developed a recommender system leveraging KGs and graph attention networks that achieved an impressive F1-score of 95%, outperforming classical machine learning and transformer-based models while demonstrating robustness against concept drift over time.

Knowledge-aware attention mechanisms have shown particular promise. [2] introduced KAN (Knowledge-aware Attention Network), which aligns entity mentions in news with KG entities and employs specialized attention modules to weigh the importance of knowledge-level relationships. This ap-

proach yielded superior performance across multiple datasets. More recently, [7] proposed KGFakeNet, which combines triplet extraction from statements with attention over GenAI-generated embeddings, achieving improvements of 4% accuracy and 5% F1-score over large language model baselines.

1.3 Multi-Modal and Cross-Domain Approaches

Recent advancements have expanded into multi-modal and cross-domain fake news detection, integrating textual, visual, and social context features with knowledge graphs. [6] introduced FKGFND, which enhances multi-modal fake news detection through fine-grained knowledge graph fusion. Concurrently, [1] developed SCCN, a multimodal semantic-enhanced attention network that utilizes co-attention mechanisms to fuse information from multiple sources, significantly boosting detection accuracy and generalization across domains.

The challenge of cross-domain generalization has been addressed through innovative architectures. [9] proposed a multi-knowledge and LLM-inspired heterogeneous graph neural network that leverages large language models for feature extraction, outperforming state-of-the-art baselines in diverse settings. These approaches demonstrate the evolving sophistication of KG-based methods in handling the complex, multi-faceted nature of fake news across different domains and modalities.

1.4 Interpretability and Commonsense Knowledge Integration

A significant trend in recent literature emphasizes model interpretability through commonsense knowledge integration. [4] developed an interpretable fake news detection method based on commonsense knowledge graphs that not only achieves accuracy up to 91% but also provides transparent reasoning by highlighting which external facts support or contradict news content. This focus on explainability addresses the critical need for trustworthy fake news detection systems that can provide justification for their classifications.

The integration of commonsense knowledge with graph matching techniques has proven particularly effective for creating models that are both accurate and interpretable. These approaches enable the identification of logical inconsistencies between news content and established commonsense knowledge, providing a robust mechanism for fake news detection that aligns with human reasoning processes.

Table 1: Key Approaches and Performance in Knowledge Graph-Based Fake News Detection

Model/Approach	Key Innovation	Performance
DEAP-FAKED [8]	NLP + KG tensor decomposition	F1: 88%, 78% (2 datasets)
GAT-based Recommender [5]	KG + Graph Attention Networks	F1: 95%, robust to concept drift
KAN [2]	Knowledge-aware attention	Outperforms SOTA on 3 datasets
KEHGNN-FD [10]	Heterogeneous GNN + KG	Superior to 7 baselines
KGFakeNet [7]	Triplet extraction + attention	+4% accuracy, +5% F1 over LLMs
FKGFND [6]	Multi-modal, fine-grained KG fusion	Outperforms prior multi-modal
KG-MFEND [3]	Multi-domain KG	Best in cross-domain benchmarks
Commonsense KG [4]	Graph matching for inter-pretability	Accuracy up to 91%

1.5 Research Gaps and Future Directions

Despite significant advancements, several research gaps remain. First, while recent approaches show improved performance, many still require extensive computational resources and large training datasets, limiting their practical deployment [8]. Second, most existing methods use simplistic knowledge integration techniques such as concatenation, failing to dynamically weigh the relevance of different knowledge elements to specific contexts. Third, the generalization of models across diverse cultural and linguistic contexts remains challenging due to variations in knowledge representation and commonsense understanding.

KNOW-NET Contributions: Our proposed KNOW-NET architecture directly addresses these gaps through:

- **Computational Efficiency:** Cross-attention mechanisms combined with pre-trained language models enable state-of-the-art performance with minimal training data (200 samples) and rapid convergence (3 epochs), addressing deployment limitations
- **Intelligent Knowledge Integration:** Dynamic attention weighting allows the model to focus on the most relevant KG entities for each specific news context, moving beyond static concatenation approaches used in prior work
- **Multi-Task Learning:** Joint optimization of fake news classification

and named entity recognition enhances feature learning and model generalization

Future research should focus on developing more efficient KG construction and updating mechanisms, exploring cross-lingual and cross-cultural knowledge integration, and enhancing model robustness against deliberately crafted misinformation. Additionally, the integration of temporal dynamics in knowledge graphs to account for evolving facts and contexts represents a promising direction for addressing the challenge of "temporal fake news" where information may be contextually accurate but temporally misaligned.

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